Supplementary Information

Identifying Sex-Specific Sub-phenotypes of Alzheimer's Disease Progression Using Longitudinal Electronic Health Records

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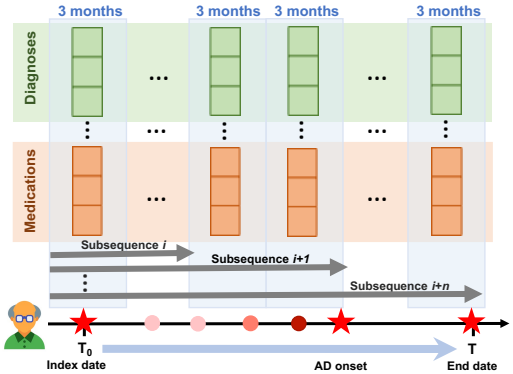
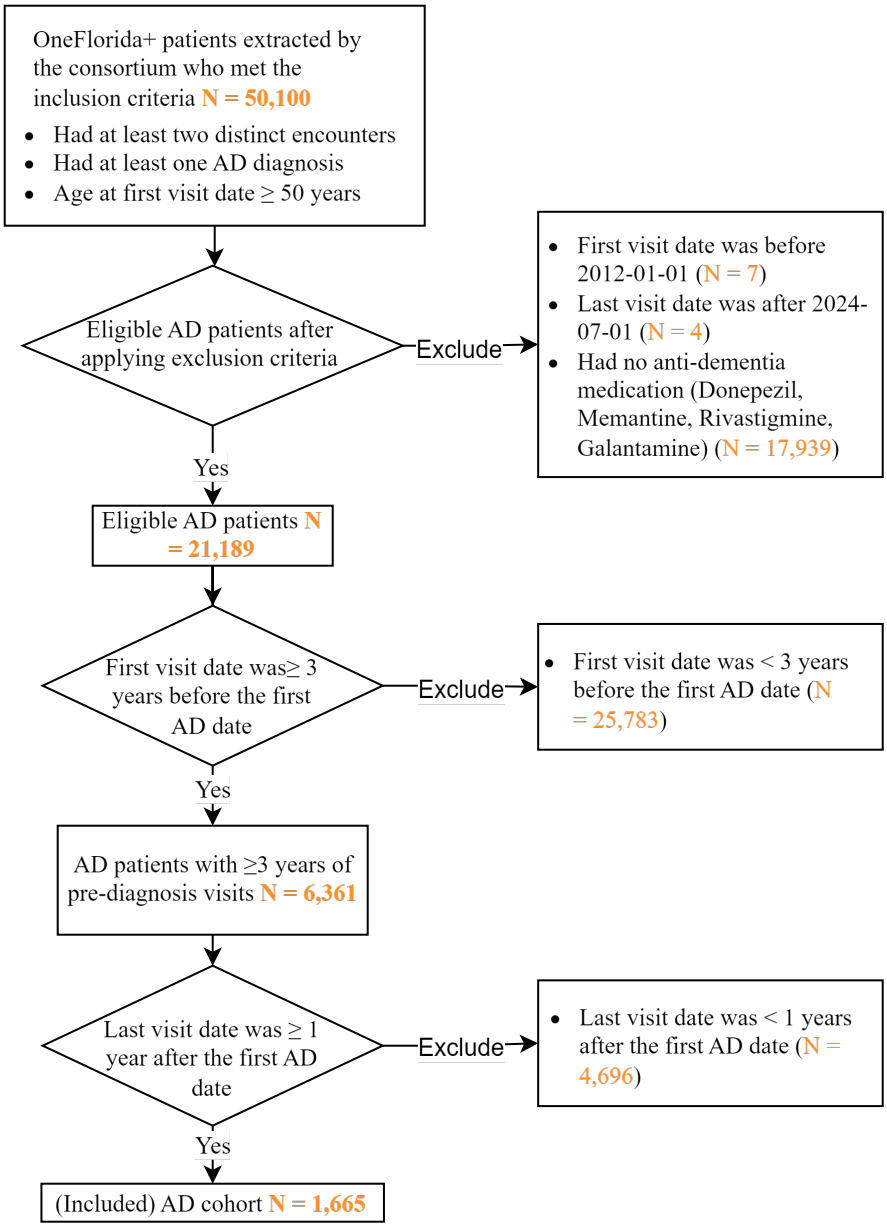
Supplementary Figures



Supplementary Figure 1. The construction of Alzheimer's Disease (AD) temporal trajectory using EHRs. Relevant EHR data for each patient was aggregated in 3-month blocks (i.e., time intervals) into a set of vectors (e.g., Diagnoses, Medications). A vector contains unique clinical variables and their frequencies within a time interval. For the invariant variables, demographic, we treated them as static features and passed them at each time interval. We then divided each patient's vectors into multiple subsequences with varying time lengths, starting from the index date, and new subsequences are created with 3-month increments (i.e., 6-month, 9-month, etc.) until the last encounter of patients within the end date.



Supplementary Figure 2. Cohort selection workflow for identifying the study population of patients with AD. The initial de-identified dataset provided by the OneFlorida+ consortium included 50,100 patients who met the predefined extraction criteria (age ≥ 50 years at the first visit, at least two distinct encounters, and at least one AD diagnosis). From this candidate cohort, we applied several filtering steps to ensure clinical consistency and sufficient longitudinal observation. Patients were excluded if their visits fell outside the study observation window or if they had no anti-dementia medications. To enable trajectory modeling, we further required each patient to have at least three years of clinical records prior to the first AD diagnosis and at least one year of follow-up after diagnosis. After applying all inclusion and exclusion criteria, the final analytical cohort consisted of 1,665 patients with AD. The number of patients retained at each filtering step is shown in the workflow.

Supplementary Tables

Supplementary Table 1. Phecodes for the three most common comorbidities of AD.

Disease Category	Neurological disorders[1–3]	Cardiovascular diseases[4,5]	Diabetes[6,7]
Phecodes	Phe_Memory loss, Phe_Altered mental status, Phe_Transient alteration of awareness, Phe_Hallucinations, Phe_Schizophrenia and other psychotic disorders, Phe_Schizophrenia, Phe_Paranoid disorders, Phe_Mood disorders, Phe_Bipolar, Phe_Depression, Phe_Major depressive disorder, Phe_Suicidal ideation, Phe_Anxiety disorders, Phe_Anxiety disorder, Phe_Phobia, Phe_Personality disorders, Phe_Eating disorder, Phe_Anorexia nervosa, Phe_Sleep disorders, Phe_Insomnia.	Phe_Essential hypertension, Phe_Hypertension, Phe_Hypertensive heart and/or renal disease, Phe_Other forms of chronic heart disease, Phe_Ill-defined descriptions and complications of heart disease, Phe_Hypertensive heart disease, Phe_Congestive heart failure, nonhypertensive, Phe_Other chronic ischemic heart disease, unspecified, Phe_Heart failure with reduced EF [Systolic or combined heart failure], Phe_Acute rheumatic heart disease	Phe_Type 2 diabetes, Phe_Diabetes mellitus, Phe_Type 2 diabetes with renal manifestations, Phe_Type 2 diabetes with neurological manifestations, Phe_Polyneuropathy in diabetes, Phe_Type 1 diabetes, Phe_Type 2 diabetes with ophthalmic manifestations, Phe_Diabetes type 2 with peripheral circulatory disorders, Phe_Type 1 diabetes with neurological manifestations, Phe_Diabetes type 1 with peripheral circulatory disorders, Phe_Type 1 diabetes with ophthalmic manifestations, Phe_Type 2 diabetes with ketoacidosis

Supplementary Table 2. Descriptive statistics on the characteristics of the study cohort.

AD Cohort, N (%)	Total AD Patients (N = 1,665)	Female AD Patients (N = 961, 57.7%)	Male AD Patients (N = 704, 42.3%)
Demographics			
Age at AD diagnosis, mean (std)	76.7 (9.2)	77.1 (9.5)	76.1 (8.8)
Disease development, mean (std)			
Follow-up durations	8.1 (1.70)	8.21 (1.68)	7.95 (1.72)
Years before AD diagnosis	5.6 (1.72)	5.59 (1.73)	5.6 (1.7)
Years after AD diagnosis	2.5 (1.55)	2.62 (1.59)	2.36 (1.48)
Years from MCI to AD	2.26 (1.70)	2.26 (1.72)	2.27 (1.69)
Conversion rate from MCI to AD	266 (16.0%)	150 (15.6%)	116 (16.5%)
Years from AD to death	2.28 (1.45)	2.39 (1.48)	2.16 (1.42)
Age at death	81.7 (8.9)	82.1 (9.4)	81.2 (8.4)
Mortality rate, N (%)	209 (12.55%)	109 (11.34%)	100 (14.20%)
Hispanic, N (%)			
Hispanic	426 (25.6%)	248 (25.8%)	178 (18.5%)
Not Hispanic	1,230 (73.9%)	707 (73.6%)	523 (54.4%)
No Hispanic information	9 (0.5%)	6 (0.6%)	3 (0.3%)
Race, N (%)			
American Indian or Alaska Native	0 (0.0%)	0 (0.0%)	0 (0.0%)
Asian	10 (0.6%)	5 (0.5%)	5 (0.7%)
Black or African American	259 (15.6%)	157 (16.3%)	102 (14.5%)
White	1,298 (77.96%)	741 (77.11%)	557 (79.12%)
Multiple race	15 (0.90%)	6 (0.62%)	9 (1.28%)
Unknown	83 (5.0%)	52 (5.4%)	31 (4.40%)
Vital Signs			
Body Mass Index (BMI)	26.5 (5.11)	26.8 (4.21)	26.3 (5.71)
Smoking status, N (%)			
Non-smoker	125 (7.5%)	81 (8.4%)	44 (4.6%)
Current smoker	15 (0.9%)	10 (1.0%)	5 (0.5%)
Ex-smoker	87 (5.2%)	37 (3.9%)	50 (5.2%)
Unknown	1,438 (86.4%)	833 (86.7%)	605 (63.0%)

Comorbidity, N (%)			
Neurological disorders	1,483 (89.1%)	876 (91.2%)	607 (86.2%)
Cardiovascular diseases	696 (41.8%)	370 (38.5%)	326 (46.3%)
Diabetes	779 (46.8%)	439 (45.7%)	340 (48.3%)

Supplementary Table 3. Descriptive statistics on the characteristics of sex-specific AD progression subsequence clusters (i.e., states)

States, N (%) [out of N = 27,328]	**C1 (N = 4,726, 17.3%)**	**C2 (N= 983, 3.6%)**	**C3 (N= 2,498, 9.1%)**	**C4 (N= 9,537, 34.9%)**	**C5 (N= 1,258, 4.6%)**	**C6 (N= 3,743, 13.7%)**	**C7 (N= 4,583, 16.8%)**
Demographics							
Age at AD diagnosis, mean (std)	77.1 (9.27)	78.1 (9.85)	77.6 (9.54)	77.2 (9.43)	75.5 (9.27)	76.1 (8.73)	77.2 (8.35)
N (%) of female AD patients with a specific state	921 (95.8%)	554 (57.7%)	866 (90.1%)	846 (88.0%)	52 (5.4%)	3 (0.3%)	1 (0.1%)
N (%) of male patients with a specific state	704 (100.0%)	18 (2.6%)	4 (0.6%)	0 (0.0%)	514 (73.0%)	681 (96.7%)	531 (75.4%)
Disease development, mean (std)							
Follow-up durations	8.1 (1.7)	8.16 (1.66)	8.23 (1.67)	8.31 (1.65)	7.98 (1.69)	7.98 (1.7)	8.17 (1.6)
Years before AD diagnosis	5.59 (1.71)	5.55 (1.72)	5.59 (1.74)	5.65 (1.74)	5.56 (1.64)	5.61 (1.7)	5.73 (1.71)
Years after AD diagnosis	2.51 (1.55)	2.61 (1.6)	2.64 (1.59)	2.66 (1.58)	2.42 (1.5)	2.37 (1.49)	2.45 (1.49)
Years from MCI to AD	2.26 (1.7)	2.14 (1.62)	2.26 (1.69)	2.29 (1.68)	2.24 (1.66)	2.26 (1.74)	2.29 (1.79)
Conversion rate from MCI to AD	45 (2.7%)	36 (2.2%)	85 (5.1%)	136 (8.2%)	36 (2.2%)	85 (5.1%)	93 (5.6%)
Years from AD to death	2.28 (1.45)	2.34 (1.47)	2.34 (1.44)	2.4 (1.5)	2.36 (1.52)	2.16 (1.43)	2.18 (1.48)
Age at death	81.7 (8.9)	82.2 (9.1)	82.1 (9.7)	82.4 (9.3)	80.3 (8.9)	81.2 (8.4)	81.5 (8.3)
Hispanic, N (%)							
Hispanic	426 (25.6%)	142 (8.5%)	225 (13.5%)	223 (13.4%)	141 (8.5%)	173 (10.4%)	132 (7.9%)
Not Hispanic	1230 (73.9%)	426 (25.6%)	639 (38.4%)	619 (37.2%)	423 (25.4%)	508 (30.5%)	398 (23.9%)
No Hispanic information	9 (0.5%)	4 (0.2%)	6 (0.4%)	4 (0.2%)	2 (0.1%)	3 (0.2%)	2 (0.1%)
Race, N (%)							
American Indian or Alaska Native	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Asian	10 (0.6%)	3 (0.2%)	4 (0.2%)	5 (0.3%)	5 (0.3%)	5 (0.3%	4 (0.2%)
Black or African American	259 (15.6%)	83 (5.0%)	139 (8.3%)	143 (8.6%)	83 (5.0%)	102 (6.1%)	86 (5.2%)
White	1,298 (77.96%)	448 (26.9%)	669 (40.2%)	645 (38.7%)	448 (26.9%)	539 (32.4%)	418 (25.1%)
Multiple race	15 (0.90%)	4 (0.2%)	6 (0.4%)	6 (0.4%)	9 (0.5%)	8 (0.5%)	5 (0.3%)
Unknown	83 (5.0%)	34(2.0%)	52 (3.1%)	47 (2.8%)	21 (1.3%)	30 (1.8%)	19 (1.1%)
Vital Signs							
Body Mass Index (BMI)	26.5 (5.1)	27.0 (4.3)	26.7 (4.1)	26.7 (4.1)	26.3 (5.7)	26.4 (5.9)	26.4 (5.9)

Smoking status, N (%)							
Non-smoker	125 (7.5%)	58 (6.0%)	76 (7.9%)	65 (6.8%)	36 (3.7%)	42 (4.4%)	36 (3.7%)
Current smoker	15 (0.9%)	7 (0.7%)	9 (0.9%)	7 (0.7%)	5 (0.5%)	5 (0.5%)	3 (0.3%)
Ex-smoker	87 (5.2%)	24 (2.5%)	36 (3.7%)	33 (3.4%)	45 (4.7%)	46 (4.8%)	34 (3.5%)
Unknown	1,438 (86.4%)	483 (50.3%)	749 (77.9%)	741 (77.1%)	480 (49.9%)	591 (61.5%)	459 (47.8%)
Comorbidity, N (%)							
Neurological disorders	1,483 (89.1%)	226 (39.5%)	521 (59.9%)	780 (92.2%)	219 (38.7%)	439 (64.2%)	470 (88.3%)
Cardiovascular diseases	696 (41.8%)	64 (11.2%)	191 (22.0%)	343 (40.5%)	105 (18.6%)	235 (34.4%)	276 (51.9%)
Diabetes	779 (46.8%)	133 (23.3%)	295 (33.9%)	403 (47.6%)	174 (30.7%)	273 (39.9%)	268 (50.4%)

Supplementary Table 4. Descriptive statistics on the characteristics of sex-specific AD progression sub-phenotypes.

Sub-phenotypes, N (%) [out of N = 1,403]	**S1 [C1→C2→C3→C4] (N = 413, 24.8%)**	**S2 [C1→C3→C4] (N= 329, 19.8%)**	**S3 [C1→C5→C6→C7] (N= 346, 20.8%)**	**S4 [C1→C6→C7] (N= 178, 10.7%)**	**S5 [C1→C5→C6] (N= 137, 8.2%)**
Demographics					
Age at AD diagnosis, mean (std)	77.2 (9.8)	76.9 (9.2)	76.2 (8.7)	77.1 (8.3)	74.8 (9.5)
Female, N (%)	406 (98.3%)	326 (99.1%)	1 (0.3%)	6 (3.4%)	2 (1.5%)
Male, N (%)	7 (1.7%)	3 (0.9%)	345 (99.7%)	172 (96.6%)	135 (98.5%)
Disease development, mean (std)					
Follow-up durations	8.3 (1.61)	8.29 (1.7)	8.24 (1.57)	8.04 (1.67)	7.39 (1.8)
Years before AD diagnosis	5.67 (1.73)	5.69 (1.76)	5.76 (1.66)	5.68 (1.8)	5.24 (1.52)
Years after AD diagnosis	2.64 (1.56)	2.6 (1.54)	2.48 (1.46)	2.37 (1.52)	2.14 (1.42)
Years from MCI to AD	2.25(1.65)	2.23(1.79)	2.29(1.8)	2.3(1.8)	2.23(1.68)
Conversion rate from MCI to AD	66 (15.9%)	53 (16.2%)	58 (16.7%)	29 (16.1%)	21 (15.3%)
Years from AD to death	2.27 (1.46)	2.31 (1.32)	2.26 (1.48)	1.93 (1.51)	2.37 (1.22)
Age at death	82.9 (9.0)	82.1 (10.1)	80.7 (8.8)	82.7 (7.1)	79.6 (8.7)
Hispanic, N (%)					
Hispanic	111 (26.9%)	84 (25.5%)	85 (24.6%)	43 (24.2%)	38 (27.7%)
Not Hispanic	293 (70.9%)	240 (72.9%)	260 (75.1%)	128 (71.9%)	97 (70.8%)
No Hispanic information	2 (0.5%)	2 (0.6%)	1 (0.3%)	1 (0.6%)	1 (0.7%)
Race, N (%)					
American Indian or Alaska Native	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Asian	3 (0.7%)	0 (0.0%)	3 (0.9%)	1 (0.6%)	1 (0.7%)
Black or African American	56 (13.6%)	60 (18.2%)	55 (15.9%)	26 (14.6%)	12 (8.8%)

White	321 (77.7%)	245 (74.5%)	279 (80.6%)	133 (74.7%)	109 (79.6%)
Multiple race	2 (0.5%)	2 (0.6%)	3 (0.9%)	2 (1.1%)	3 (2.2%)
Unknown	124 (30.0%)	89 (27.1%)	122 (35.3%)	56 (31.5%)	37 (27.0%)
Vital Signs					
Height, inch	62.7 (4.8)	67.1 (57.9)	69.0 (3.1)	69.0 (2.9)	70.8 (17.6)
Weight, lbs.	152.8 (32.6)	154.3 (34.4)	193.0 (40.6)	189.1 (33.8)	182.2 (36.4)
Diastolic Blood Pressure, mmHg	72.6 (11.3)	73.3 (12.2)	76.2 (11.6)	75.7 (12.3)	74.8 (10.5)
Systolic Blood Pressure, mmHg	135.8 (21.6)	135.4 (21.4)	134.6 (20.4)	133.2 (17.6)	133.6 (22.3)
Body Mass Index (BMI)	26.9 (5.1)	27.0 (5.8)	27.9 (4.3)	27.7 (4.6)	27.9 (4.3)
Smoking status, N (%)					
Non-smoker	26 (6.3%)	11 (3.3%)	14 (4.0%)	5 (2.8%)	2 (1.5%)
Current smoker	0 (0.0%)	0 (0.0%)	3 (0.9%)	1 (0.6%)	0 (0.0%)
Ex-smoker	6 (1.5%)	6 (1.8%)	17 (4.9%)	7 (3.9%)	5 (3.6%)
Unknown	144 (34.9%)	109 (33.1%)	152 (43.9%)	66 (37.1%)	37 (27.0%)
Comorbidity, N (%)					
Neurological disorders	149 (36.1%)	113 (34.3%)	143 (41.3%)	70 (39.3%)	37 (27.0%)
Cardiovascular diseases	68 (16.5%)	65 (19.8%)	78 (22.5%)	34 (19.1%)	21 (15.3%)
Diabetes	171 (41.4%)	170 (51.7%)	178 (51.4%)	85 (47.8%)	50 (36.5%)

Supplementary Table 5. Consistency of clusters and subphenotypes across different latent dimensions (Baseline: latent dim: 16, distance metric: Euclidean, linkage method: Ward).

Latent dim	AMI of Clusters	ARI of Clusters	AMI of Subtypes	ARI of Subtypes
8	0.789	0.725	0.695	0.687
32	0.763	0.758	0.701	0.697
64	0.754	0.736	0.692	0.681

Supplementary Table 6. Consistency of clusters and subphenotypes across different linkage methods (Baseline: latent dim: 16, distance metric: Euclidean, linkage method: Ward).

Linkage	AMI of Clusters	ARI of Clusters	AMI of Subtypes	ARI of Subtypes
Complete	0.746	0.721	0.681	0.632
Average	0.856	0.812	0.711	0.697
Centroid	0.792	0.731	0.712	0.618

Supplementary Table 7. Consistency of clusters and subphenotypes across different distance metrics (Baseline: latent dim: 16, distance metric: Euclidean, linkage method: Average).

Linkage	AMI of Clusters	ARI of Clusters	AMI of Subtypes	ARI of Subtypes
Euclidean	0.856	0.812	0.711	0.697
Cosine	0.508	0.373	0.464	0.381
Correlation	0.926	0.907	0.854	0.837

Supplementary Table 8. Cox proportional hazards models results.

	Patients with Essential Hypertension		**Patients with Hyperlipidemia**		**Patients with Hypertension**	
Variable	**HR (95% CI)**	**Log-rank p**	**HR (95% CI)**	**Log-rank p**	**HR (95% CI)**	**Log-rank p**
Male- vs. Female-dominant subphenotypes	1.35 (1.05–2.16)	<0.001	1.21 (0.92–2.05)	<0.001	1.54 (1.17–2.39)	<0.001
Covariates						
Age	0.99 (0.98–1.00)	0.02	0.99 (0.98–1.00)	<0.005	0.99 (0.98–1.00)	0.19
Hispanic	0.94 (0.79–1.11)	0.44	1.14 (0.95–1.35)	0.15	1.03 (0.81–1.29)	0.83
Race: Black (ref = White)	1.11 (0.81–1.53)	0.51	0.98 (0.71–1.37)	0.92	0.83 (0.55–1.27)	0.39
Race: Other (ref = White)	1.16 (0.96–1.40)	0.12	1.11 (0.91–1.35)	0.31	1.09 (0.87–1.37)	0.46
No. encounters	1.00 (1.00–1.00)	<0.001	1.00 (1.00–1.00)	0.01	1.00 (1.00–1.00)	0.17
Model statistics						
N (events / censored)	1,403 (1,314 / 70)		1,403 (1,201 / 202)		1,403 (1,294 / 109)	
PH assumption	Pass		Pass		Pass	

Supplementary Table 9. Hyperparameter settings

Hyperparameter	**Settings**
Framework	Python (v3.12), PyTorch (v2.10.0, CUDA 11.8)
Hidden layers	[256, 128, 64, 1]
Subsequence window	6 months
Input features	Demographics + Phenotypes
Batch size	64
Max epochs	500
Learning rate	1e-3
Dropout	0.2
Weight decay	1e-3
Early stopping patience	20
Classification loss weight	0.5
Reconstruction loss weight	0.5
Decision threshold	0.5

Supplementary References

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